

Qatar-4b

R-2376

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-4_230104 · created 2026-08-20T00:48:49+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459948.6403 +/- 0.0042 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.106 +/- 0.043
Transit depth (Rp/Rs)^2	1.11 +/- 0.91 [%]
Orbital Inclination (inc)	85.39 +/- 2.13
Airmass coefficient 1 (a1)	0.934 +/- 0.037
Airmass coefficient 2 (a2)	0.055 +/- 0.032
Scatter in the residuals of the lightcurve fit is	3.93 %
Best Comparison Star	#1 - [600, 450]
Optimal Aperture	3.22
Optimal Annulus	5.35
Transit Duration (day)	0.072 +/- 0.018

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.106 $\pm$ 0.043 vs 0.138 (deviation 0.74 σ)
Duration fit vs published	0.072 d vs 0.0892 d (deviation 0.95 σ)
Mid-time O-C	-1.1 min
Depth SNR	2.5
Residual scatter	3.93 %

Light curve

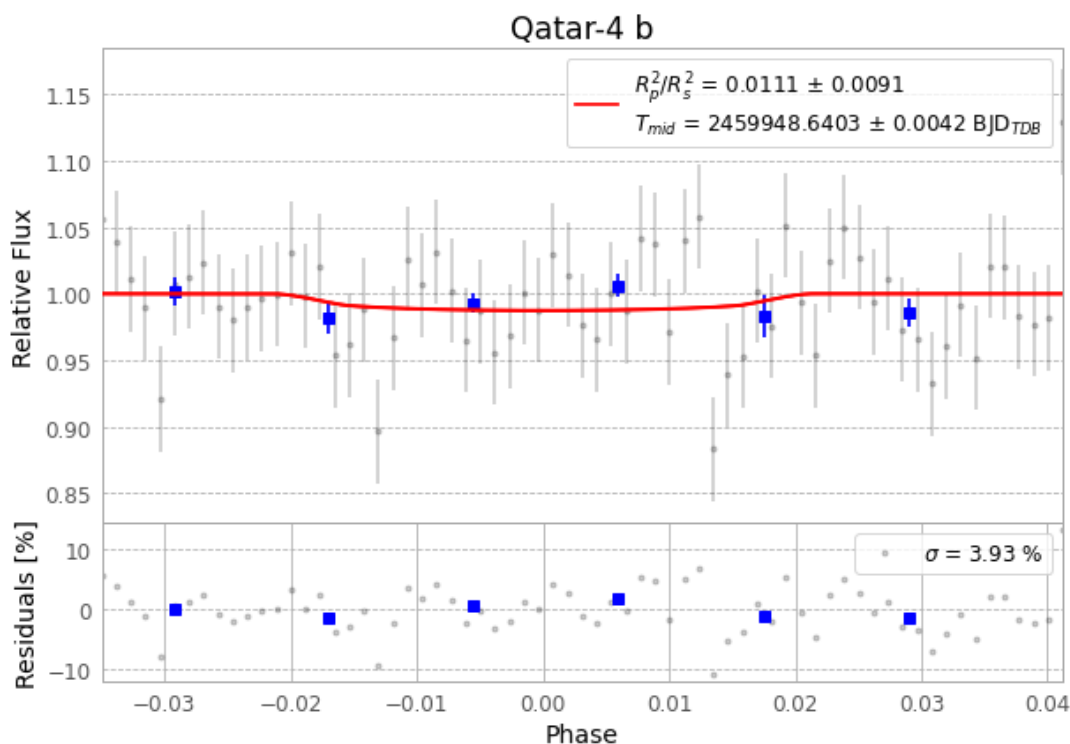


Figure 1 — FinalLightCurve_Qatar-4 b_2023-01-03.png (SHA-256 7189f3772200c304...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [458, 268], 2 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 9bb295a6b8f168eb...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

71 FITS frames (47 MB), Qatar-4230104014815.FITS → Qatar-4230104051815.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-4-230104.log (4223ff9d1849...), FinalLightCurve_Qatar-4 b_2023-01-03.png (7189f3772200...), FinalLightCurve_Qatar-4 b_2023-01-03.csv (3499209d9d96...)

Compute resources

Wall time 107.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-20 00:48Z.